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Assignment #2

1. What is network architecture?

Network architecture refers to the design and structure of a computer network. It encompasses the physical and logical layout of the network, including the hardware components (like routers, switches, and servers), the software protocols used for communication, and the overall framework that dictates how data is transmitted and received across the network. Network architecture can be categorized into different types, such as client-server, peer-to-peer, and hybrid models, and it plays a crucial role in determining the network's performance, scalability, and security.

2. What is a network domain?

A network domain is a logical grouping of network resources, such as computers, devices, and users, that are managed as a single entity. It typically refers to a collection of devices that share a common set of rules and policies, often governed by a central authority, such as a domain controller in a Windows environment. Domains help in organizing and managing network resources, providing a way to implement security policies, user authentication, and resource sharing within a defined boundary.

3. What is a network application? Give 10 example of network applications

A network application is a software program that operates over a network, allowing users to communicate, share resources, or access services remotely. These applications rely on network protocols to transmit data between clients and servers. Here are 10 examples of network applications:

- 1. Web Browsers:** (Google Chrome, Mozilla Firefox)
- 2. Email Clients:** (Microsoft Outlook, Mozilla Thunderbird)
- 3. File Transfer Protocol (FTP) Clients:** (FileZilla, WinSCP)
- 4. Instant Messaging Applications:** (Slack, Microsoft Teams)
- 5. Video Conferencing Tools:** (Zoom, Microsoft Teams)
- 6. Remote Desktop Applications:** (TeamViewer, Remote Desktop Protocol)
- 7. Online Gaming Platforms:** (Steam, Xbox Live)
- 8. Social Media Platforms:** (Facebook, Twitter)
- 9. Cloud Storage Services:** (Google Drive, Dropbox)
- 10. VoIP Services:** (Skype, WhatsApp)

4. What is OSI and TCP/IP model?

The OSI (Open Systems Interconnection) model is a conceptual framework used to understand and implement network communication. It divides the networking process into seven layers, each with specific functions:

1. **Physical Layer:** Deals with the physical connection between devices.
2. **Data Link Layer:** Manages node-to-node data transfer and error detection.
3. **Network Layer:** Handles routing and forwarding of data packets.
4. **Transport Layer:** Ensures reliable data transfer and error recovery.
5. **Session Layer:** Manages sessions and controls the dialog between applications.
6. **Presentation Layer:** Translates data formats and encrypts/decrypts data.
7. **Application Layer:** Provides network services to end-user applications.

The TCP/IP (Transmission Control Protocol/Internet Protocol) model is a more simplified framework that consists of four layers:

1. **Link Layer:** Corresponds to the OSI's Physical and Data Link layers.
2. **Internet Layer:** Similar to the OSI's Network layer, it handles packet routing.
3. **Transport Layer:** Equivalent to the OSI's Transport layer, it provides communication services.
4. **Application Layer:** Combines the OSI's Session, Presentation, and Application layers, providing services to user applications.

5. What is TCP and UDP?

TCP (Transmission Control Protocol) and **UDP (User Datagram Protocol)** are two core protocols of the Transport layer in the TCP/IP model.

- **TCP:**
 - Connection-oriented protocol, meaning it establishes a connection before data transmission.
 - Ensures reliable data transfer through error checking, acknowledgments, and retransmissions.
 - Data is sent in a stream of bytes, and packets are reassembled in the correct order.
 - Commonly used for applications where reliability is crucial, such as web browsing (HTTP/HTTPS), email (SMTP), and file transfers (FTP).

- **UDP:**

- Connectionless protocol, meaning it does not establish a connection before sending data.
- Does not guarantee reliable delivery, order, or error checking, making it faster but less reliable than TCP.
- Data is sent in discrete packets called datagrams, which may arrive out of order or not at all.
- Commonly used for applications where speed is more important than reliability, such as video streaming, online gaming, and VoIP.